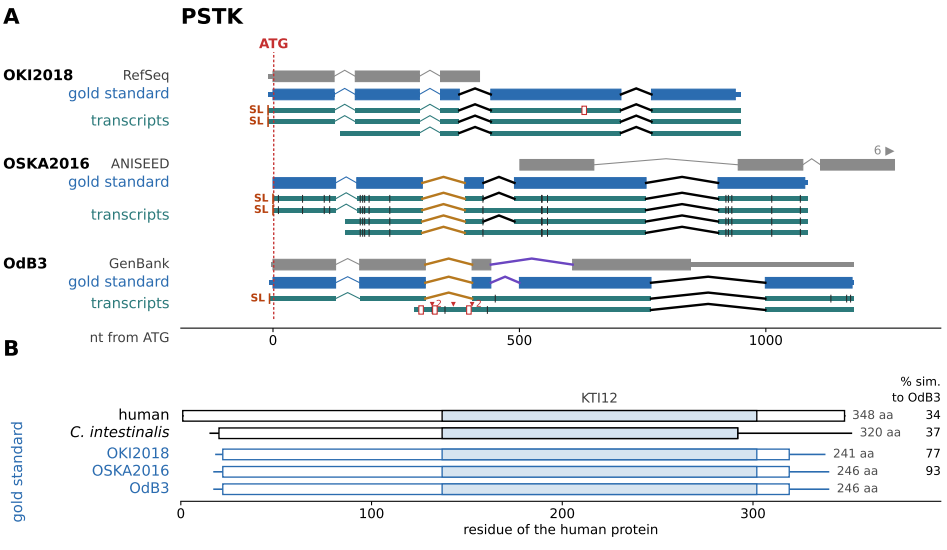


OKI2018 RefSeq XP_081642556.1: first three exons only; reads on into the third intron (110 codons, 98 of them as in the gold standard). Gold standard: third and fourth introns with splice sites outside the intron classes (TG-AG, AG-AA), as in the three supporting transcripts. Transcripts (top to bottom): TRINITY_DN16768_c0_g1_i1 (SL), TRINITY_DN16768_c0_g1_i3 (SL), TRINITY_DN16768_c0_g1_i4.

OSKA2016 ANISEED S10.g09825.01.t: starts in the fourth exon and continues past the stop codon into seven exons downstream (604 codons). Gold standard: third and fourth introns with splice sites outside the intron classes (AA-AG, TT-GT), as in the supporting transcripts; the third intron is retained in two of them. Transcripts (top to bottom): comp20361_c0_seq1 (SL), comp20361_c0_seq5 (SL), comp20361_c0_seq6, comp20361_c0_seq4.

OdB3 GenBank CBY23135.1: same first three exons; other first site of the fourth exon, and no last exon (178 codons). Gold standard: exon chain set by the curator: the GenBank exons 1 to 3, the acceptor of the third intron (GA-AG; retained in both ESTs) from the prediction, and the last intron of the ESTs (AT-AG). Transcripts (top to bottom): KT0AAA259YN02RM1 (SL), KT0AAA109YN17RM1. Also starting with the SL, not drawn: FP831546, FP830122.

proteins human Q8IV42; C. intestinalis XP_002120985.2.

C Equivalent placements of introns

Third intron

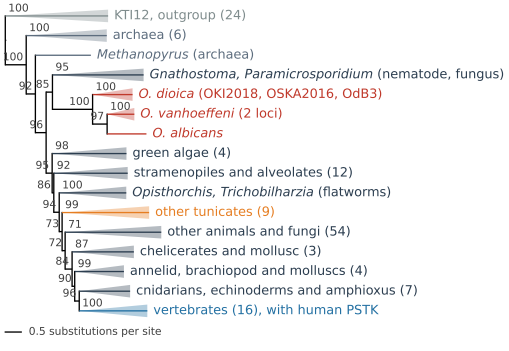
OKI2018	cgaataaTGCAGGG … TTATCAGtcttt	**TG-AG**	GC-GT	
KUM-M3-7f	cgaataaTGCAGGG … TTATCAGtcttt	**TG-AG**	GC-GT	
	R I I F			
OSKA2016	cgaatagAAGGGTT … CAACGAGttttc	AG-CG	GA-GA	**AA-AG**
AOM-5-5f	cgaatagAAGGGTT … CAACGAGttttc	AG-CG	GA-GA	**AA-AG**
OdB3	cgaatagGAGGGTT … CAACGAGttttc	AG-CG	GG-GA	**GA-AG**
Bar2_p4	cgaatagGAGGGTT … CAACGAGttttc	AG-CG	GG-GA	**GA-AG**
	R I V F			

Fourth intron

OKI2018	cacttagTTTGGGT … TTTTGAAAGtagaa	AG-AA	GT-AA	**TT-AG**	TT-GT
KUM-M3-7f	cacttagTTTGGGT … TTTTGAAAGtagaa	AG-AA	GT-AA	**TT-AG**	TT-GT
OSKA2016	catttagTTTGGGT … GTGTGACAGtagaa	AG-AC	GT-CA	**TT-AG**	TT-GT
AOM-5-5f	catttagTTTGGGT … GTATGACAGtagaa	AG-AC	GT-CA	**TT-AG**	TT-GT
OdB3	catttagATCGGGT … CTGTGTCAGtggaa	AG-TC	GA-CA	**AT-AG**	
Bar2_p4	catttagATCGGGT … CTGTGTCAGtggaa	AG-TC	GA-CA	**AT-AG**	
	H L V E				

Gene strand, cased for the placement named by the gold-standard script (▲): lowercase grey, exon; uppercase, intron. Shaded: bases found at both ends, across which the intron can be moved without changing the spliced sequence. Rows aligned on the codons of the placement on an AG acceptor (phase 1, blue), light lines between codons; exon flanks trimmed to whole codons. Letters: amino acids under the middle base of each codon of that placement (the codon split by the intron on the side holding two of its bases), one line under the rows that translate alike. Right: dinucleotides of each placement, from left to right. Rows in italics: test assemblies (other individuals of the same species), exons carried from the gold standard.

D Simplified tree of PSTK, more eukaryotes and archaea (PMSF)



Nodes supported by less than 50% collapsed; ultrafast bootstrap supports of 70% or more shown. Wedges: clades of sequences of one group, from their ancestor to the deepest sequence; “unresolved”: sequences of the other groups in one polytomy, drawn as one tip. Sequences added to the tree of Figure TREES: see Methods.

exon: coding (thick), UTR (thin) — public model — gold-standard model — coding sequence in another frame

start codon of the gold-standard model (x axis: nucleotides from it, gene orientation) — ◂ n number of exons beyond the drawn region

a b public models of the same row, as named in the paragraph

transcript aligned to the assembly (exons) — bases of the transcript missing from the assembly (number when more than one)

bases of the assembly missing from the transcript — mismatches — SL transcript that starts with the spliced leader

intron splice sites (drawn length unchanged): GT-AG (colour of the model) — GA-AG — GG-AG — other site

Pfam domain of the first reference — residues not aligned to the first reference

92 % similar residues to the OdB3 protein (BLOSUM62 > 0; global alignment, aligned part) — human: UniProt; Ciona: RefSeq